

DISCOVERING HIDDEN RELATIONS AMONG TRIANGLE INVARIANTS

TRIANGLE-RELATIONS PROJECT

ABSTRACT. We describe a numerical method for discovering previously unknown functional relations among scalar quantities derived from a triangle, in the spirit of Euler’s classical relation between the circumradius, the inradius, and the distance between the incenter and circumcenter. Because a triangle has three degrees of freedom up to rigid motion, a relation among four or more derived scalars is guaranteed by dimension counting alone and is therefore uninformative; a relation among exactly three scalars, by contrast, is a genuine and generically unexpected algebraic coincidence. We detect such coincidences by training a bottleneck autoencoder on sampled triangle data and comparing its reconstruction error against a column-permuted null. Applied to the circumradius, inradius, and incenter-to-circumcenter distance, the method flags exactly this triple as functionally dependent, with no prior knowledge of Euler’s relation.

1. SETUP: THE MODULI SPACE OF TRIANGLES

A triangle is specified by three points in $\mathbb{R}^2$, i.e. six real parameters. Every quantity we consider (area, perimeter, circumradius R , inradius r , distances between derived points, ...) is built purely from the vertex coordinates and is invariant under rigid motions: translating or rotating the triangle does not change its area, its inradius, or the distance between its incenter and circumcenter. Translations (2 parameters) and rotations (1 parameter) form a 3-dimensional group acting on the 6-dimensional space of vertex triples; quotienting it out leaves a 3-dimensional moduli space $\mathcal{M}$ of triangle shapes up to congruence (reflection is a further discrete symmetry and does not change this dimension count). Concretely, $\mathcal{M}$ can be coordinatized by, e.g., the three side lengths (a, b, c) subject to the triangle inequality.

Every derived scalar quantity is a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$. Choosing k such quantities gives a smooth map

$$F = (f_1, \dots, f_k) : \mathcal{M} \rightarrow \mathbb{R}^k.$$

Recall that a (smooth) m -dimensional submanifold of $\mathbb{R}^n$ is a subset that, near each of its points, agrees with the graph of a smooth function expressing $n - m$ of the coordinates in terms of the other m ; this is what lets us speak of $\mathcal{M}$ itself, and of the image of F , as having a well-defined dimension even though neither is simply a coordinate subspace of $\mathbb{R}^k$. We call a property of such a map *generic* if it holds for every choice of F outside a closed, measure-zero exceptional set — informally, for “almost every” choice of derived quantities, and stably so under small perturbations of them.

2. WHY RELATIONS AMONG FOUR OR MORE QUANTITIES ARE NOT INTERESTING

If $k > \dim \mathcal{M} = 3$, a generic smooth map from a 3-manifold into $\mathbb{R}^k$ has, as its image, an (at most) 3-dimensional submanifold of $\mathbb{R}^k$ — this is just the statement that embedding a 3-dimensional space into a higher-dimensional ambient space produces $k - 3$ constraint equations for free, regardless of which k functions were chosen, as long as they are not specially degenerate. Concretely, this image

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is cut out locally by $k - 3$ independent equations $G_i(f_1, \dots, f_k) = 0$. So finding *some* relation among four or more triangle invariants is guaranteed by dimension counting alone. It says nothing specific about triangle geometry — it would happen for essentially any four smooth functions on any 3-dimensional space of objects.

3. WHY A RELATION AMONG EXACTLY THREE QUANTITIES IS SPECIAL

If $k = \dim \mathcal{M} = 3$ exactly, the situation is qualitatively different. Recall that the Jacobian $DF(p)$ of a differentiable map $F = (f_1, f_2, f_3) : \mathcal{M} \rightarrow \mathbb{R}^3$ at a point p is the 3×3 matrix of partial derivatives $(\partial f_i / \partial x_j)$ in local coordinates x_j on $\mathcal{M}$; its rank measures how many of the f_i vary independently of one another near p . Recall also that F is a *local diffeomorphism* at p if it restricts to a smooth bijection, with smooth inverse, from some neighborhood of p onto some neighborhood of $F(p)$ — by the inverse function theorem, exactly when $DF(p)$ is invertible.

A generic smooth map $F : \mathcal{M} \rightarrow \mathbb{R}^3$ is, at a generic point, a local diffeomorphism: its Jacobian DF has full rank 3, and as the triangle shape ranges over an open subset of $\mathcal{M}$, the triple (f_1, f_2, f_3) sweeps out a full 3-dimensional *open region* of $\mathbb{R}^3$ — a solid blob, not a surface. This is again a genericity statement: for an arbitrary hand-picked triple of triangle invariants, there is no a priori reason for DF to be rank-deficient anywhere.

An *exact* relation $G(f_1, f_2, f_3) = 0$ holding on all of $\mathcal{M}$ is therefore equivalent to DF having rank ≤ 2 everywhere: the image collapses onto a 2-dimensional surface embedded in $\mathbb{R}^3$, instead of filling out a full 3-dimensional region. That is not generic — it is a genuine algebraic coincidence, exactly the kind of fact that constitutes a classical theorem. Euler's relation

$$d^2 = R^2 - 2Rr$$

(where d is the distance between the incenter and circumcenter) is exactly a statement that the Jacobian of (R, r, d) as functions of triangle shape has rank ≤ 2 everywhere, collapsing three ostensibly independent quantities onto a single surface. This is the precise sense in which searching for relations among exactly three derived scalars — and not four — is the search for something structurally new.

4. DETECTING RANK-DEFICIENCY WITH A BOTTLENECK AUTOENCODER

Given a candidate triple (f_1, f_2, f_3) , we want a numerical test for whether the induced map $F : \mathcal{M} \rightarrow \mathbb{R}^3$ has full rank 3 generically (data fills a 3D region) or is everywhere rank ≤ 2 (data is confined to a 2D surface). We sample many random triangles — many points of $\mathcal{M}$ — and evaluate (f_1, f_2, f_3) on each, producing a point cloud $X \subset \mathbb{R}^3$.

An autoencoder with a 2-dimensional latent bottleneck, an encoder $e : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and decoder $d : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ trained to minimize $\mathbb{E} \|d(e(x)) - x\|^2$ over the sample, is exactly searching for the best rank-2 (nonlinear) approximation to the data:

- If the data truly lies on a 2-dimensional surface (an exact relation holds), a 2-dimensional latent code is not actually a bottleneck for it — a sufficiently expressive encoder/decoder pair can in principle drive the reconstruction error to (near) zero, since the encoder just needs to learn a local chart of the surface and the decoder its inverse.
- If the data genuinely fills an open 3-dimensional region, no continuous map that factors through a 2-dimensional intermediate space can reconstruct it without loss: an entire dimension worth of variation is necessarily discarded. Reconstruction error is then bounded away from zero, on the order of the data's own extent in the discarded direction.

This gives a clean signature: near-zero reconstruction error with a 2-dimensional bottleneck is evidence for an (exact or near-exact) relation among the three chosen quantities; error comparable to the data’s own scale is consistent with the generic, full-rank, no-relation case.

4.1. A precise statement. We can make this dichotomy precise. Let P denote the law of $(f_1, f_2, f_3)(T)$ for a random triangle T , and write $X = \text{supp}(P) \subset \mathbb{R}^3$ for the support of P (the smallest closed set carrying all of its probability mass).

By a *two-layer perceptron of hidden width h* we mean a function $N : \mathbb{R}^p \rightarrow \mathbb{R}^q$ of the form

$$N(x) = W_2 \sigma(W_1 x + b_1) + b_2,$$

where $W_1 \in \mathbb{R}^{h \times p}$, $b_1 \in \mathbb{R}^h$, $W_2 \in \mathbb{R}^{q \times h}$, $b_2 \in \mathbb{R}^q$ are the trainable weights and biases (“two-layer” for the two weight matrices W_1, W_2 ; this is what our encoders and decoders are individually, before being composed into the single autoencoder network of §4), and σ is a fixed nonlinear activation applied componentwise — here $\sigma = \tanh$. Recall that a function g is K -Lipschitz if $\|g(x) - g(y)\| \leq K\|x - y\|$ for all x, y ; since $\tanh$ itself is 1-Lipschitz, every two-layer perceptron is Lipschitz, with constant controlled by the operator norms of W_1 and W_2 . For a fixed hidden width h , let $\mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ denote the class of all two-layer perceptrons $\mathbb{R}^p \rightarrow \mathbb{R}^q$ of that width. We train an encoder $e \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2)$ and a decoder $d \in \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ to minimize the reconstruction risk

$$L(e, d) = \mathbb{E}_{x \sim P} \|d(e(x)) - x\|^2.$$

Proposition 1 (Sufficiency). *Recall that a subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Heine–Borel) — for us, X is compact whenever the random triangles are sampled from a bounded region, as they are in practice. If X is compact and admits continuous maps $\varphi : X \rightarrow \mathbb{R}^2$ and $\psi : \varphi(X) \rightarrow \mathbb{R}^3$ with $\psi(\varphi(x)) = x$ for all $x \in X$ — in particular, whenever an exact relation $G(f_1, f_2, f_3) = 0$ confines X to a 2-dimensional set — then for every $\varepsilon > 0$ there is a width h and a pair $(e, d) \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2) \times \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ with $L(e, d) < \varepsilon$.*

Proof sketch. Recall the universal approximation theorem (Cybenko, 1989; Hornik, 1991): for any continuous $\varphi : K \rightarrow \mathbb{R}^q$ on a compact set K and any $\delta > 0$, there is a hidden width h and a two-layer perceptron $N \in \mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ with $\sup_{x \in K} \|N(x) - \varphi(x)\| < \delta$. Applying this to φ on X and to ψ on $\varphi(X)$ gives encoders/decoders e, d with $\sup_{x \in X} \|d(e(x)) - x\| \rightarrow 0$ as $\delta \rightarrow 0$, hence $L(e, d) \rightarrow 0$. $\square$

Proposition 2 (Necessity, at bounded capacity). *Fix $K > 0$. Recall that a probability measure P on $\mathbb{R}^n$ is absolutely continuous (with respect to Lebesgue measure) if it assigns zero probability to every set of Lebesgue measure zero, i.e. it has a density; this is the standard way of saying “ P does not secretly concentrate on some lower-dimensional set.” Suppose X contains an open ball $B \subset \mathbb{R}^3$ on which P is absolutely continuous with $P(B) > 0$. Then there is a constant $c = c(P|_B, K) > 0$ such that $L(e, d) \geq c$ for every pair of K -Lipschitz maps $e : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $d : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ — in particular for every $(e, d) \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2) \times \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$, of any width h , whose realized weights keep the Lipschitz constant below K .*

Proof sketch. Recall that the Hausdorff dimension of a set $S \subset \mathbb{R}^n$ generalizes the usual notion of dimension to arbitrary (possibly non-smooth) sets: informally, it is the value α such that covering S by balls of radius ε requires on the order of $\varepsilon^{-\alpha}$ of them as $\varepsilon \rightarrow 0$; it agrees with the usual dimension on smooth submanifolds, and a standard fact is that Lipschitz maps do not increase it. Since d is K -Lipschitz, its image $d(\mathbb{R}^2)$ therefore has Hausdorff dimension at most 2, and a subset of $\mathbb{R}^3$ of Hausdorff dimension less than 3 automatically has (ordinary, 3-dimensional) Lebesgue measure zero, i.e. is *Lebesgue-null*. As P is absolutely continuous on B with $P(B) > 0$, it assigns zero mass to this null set, so $\text{dist}(x, d(\mathbb{R}^2)) > 0$ for $P|_B$ -almost every x ; a routine compactness argument

over K -Lipschitz maps then gives a strictly positive lower bound c on $\mathbb{E}_{x \sim P|_B} \text{dist}(x, d(\mathbb{R}^2))^2$. Since $d(e(\mathbb{R}^3)) \subseteq d(\mathbb{R}^2)$, we have $\|d(e(x)) - x\| \geq \text{dist}(x, d(\mathbb{R}^2))$ pointwise, and the same bound applies to $L(e, d)$. $\square$

Remark 3. *The capacity bound K is not a mere technicality. Recall Brouwer’s invariance-of-domain theorem: a continuous injective map between open subsets of $\mathbb{R}^n$ is necessarily an open map (its image is open); in particular, no continuous injection of $\mathbb{R}^2$ into $\mathbb{R}^3$ can have open image, so an exact space-filling reconstruction is topologically impossible regardless of capacity. What Brouwer’s theorem does not rule out is approximate space-filling: an unconstrained sequence of increasingly ill-conditioned (large-Lipschitz-constant) maps can approach such behavior arbitrarily closely without ever achieving it, which would erode the uniform lower bound c as $K \rightarrow \infty$. In practice this is not a loophole: the ℓ_2 weight regularization and early stopping used when training the autoencoder keep the realized Lipschitz constant controlled, which is exactly why the dichotomy above is what we observe empirically.*

Together, Propositions 1 and 2 say that, at any fixed, reasonably regularized network capacity, whether reconstruction error tends to 0 or stays bounded away from it is not a matter of degree: it is decided entirely by whether the sampled data is confined to a 2-dimensional subset of $\mathbb{R}^3$, i.e. by whether f_1, f_2, f_3 satisfy an exact functional relation on the region sampled. This is the theoretical content behind the empirical detection rule used throughout this paper.

5. CALIBRATING “NEAR ZERO” WITH A PERMUTATION NULL

An absolute error threshold is not meaningful across different triples of scalars: they differ in units, scale, and marginal shape, and — since all derived scalars are ultimately functions of the same three underlying triangle parameters — some triples are mildly correlated even without an exact relation. What is needed is a triple-specific baseline for “no relation.”

We construct a null sample X' by independently permuting each of the three columns of X . This exactly preserves each f_i ’s own one-dimensional marginal distribution while completely destroying the joint dependency between the three: each row of X' combines values from unrelated triangle instances. Any real functional relation is annihilated by this shuffle. Training the same autoencoder architecture on X' (repeated a few times) estimates the reconstruction error one should expect for three quantities with these particular marginals but *no* shared structure. Writing L_{real} for the reconstruction risk on the real data and $\mu_{\text{null}}, \sigma_{\text{null}}$ for the mean and standard deviation of the reconstruction risk over the shuffled repeats, we compare the two via the standard z -score

$$z = \frac{\mu_{\text{null}} - L_{\text{real}}}{\sigma_{\text{null}}},$$

i.e. the number of null standard deviations by which the real error falls below the null mean, or equivalently via the ratio $L_{\text{real}}/\mu_{\text{null}}$. Both give a self-calibrated significance measure that is robust to the units and distribution of whichever three scalars are being tested: a large z (or a small ratio) — real error far below the null — is the fingerprint of an unexpected relation among exactly three quantities.

6. THE FULL PIPELINE

- (1) Enumerate candidate triples of derived scalar quantities.
- (2) For each triple, sample random triangles once, train a 2-bottleneck autoencoder, and compare its held-out reconstruction error to a column-shuffled null (several repeats).

- (3) Rank triples by how much better the real data compresses than its null counterpart (small ratio = strong candidate).

The autoencoder is used purely as a screening tool here: it tells us *whether* a triple of quantities is suspiciously dependent, not what the relation is. The `verify_euler_relation` module (in `triangle_relations/discovery`) runs this exact pipeline on (R, r, d) as a worked example, confirming it correctly flags this triple — Euler’s relation — with no prior knowledge of the formula.